Resampling Techniques

Problem 18: Bayesian Linear Regression

Goal:

- Implement Standard Gibbs vs. Blocked Gibbs Sampling
- Evaluate convergence using Trace Plots and ACF
- Analyze the impact of feature correlation on mixing

Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

- Dataset provided with high correlation between x_1 and x_2

Prior Distributions:

- $\boldsymbol{\beta} \sim N(\mu_0, \Lambda_0^{-1})$
- $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$

Defining the Blocks

In Gibbs sampling, we partition the set of all parameters Θ into disjoint subsets called **Blocks**.

The Partitioning

For our Linear Regression model, $\Theta = \{\beta_0, \beta_1, \beta_2, \sigma^2\}$.

- **Standard Gibbs**: 4 Blocks

$$B_1 = \{\beta_0\}, B_2 = \{\beta_1\}, B_3 = \{\beta_2\}, B_4 = \{\sigma^2\}$$

- **Blocked Gibbs**: 2 Blocks

$$B_1 = \{\beta_0, \beta_1, \beta_2\}, B_2 = \{\sigma^2\}$$

Key Idea: Larger blocks reduce autocorrelation but require sampling from Multivariate distributions.

How to Update Blocks (The Algorithm)

The update rule is: **Sample each block conditioned on the current values of all other blocks.**

For the 2-block version (Blocked Gibbs):

1. **Update Block 1 (β):** Keep σ^2 fixed at its last value. Draw a new β from:

$$P(\boldsymbol{\beta} \mid \sigma^2_{old}, \mathbf{y})$$

2. **Update Block 2 (σ^2):** Keep the *new* $\boldsymbol{\beta}$ fixed. Draw a new σ^2 from:

$$P(\sigma^2 \mid \boldsymbol{\beta}_{new}, \mathbf{y})$$

Iteration Rule

One full iteration consists of updating every block exactly once.

The Mathematical Updates

Step 1: Sampling the $\boldsymbol{\beta}$ Block

Using the current σ^2, sample from the Multivariate Normal:

$$\boldsymbol{\beta} \sim \mathcal{N}\left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1}\right)$$

Step 2: Sampling the σ^2 Block

Using the current $\boldsymbol{\beta}$, sample from the Inverse-Gamma:

$$a_n = n/2, \quad b_n = \frac{1}{2}(y - X\boldsymbol{\beta})^T (y - X\boldsymbol{\beta})$$

$$\sigma^2 \sim \text{Inv-Gamma}(a_n, b_n)$$

Implementation: The Conditionals

Standard Gibbs (Scalar Updates)

$$\beta_j | \beta_{=j}, \sigma^2 \sim \mathcal{N}\left(\frac{\sum x_{ij}(y_i - \sum_{k \neq j} x_{ik}\beta_k)}{\sum x_{ij}^2}, \frac{\sigma^2}{\sum x_{ij}^2} \right)$$

Blocked Gibbs (Vector Update)

$$\boldsymbol{\beta} | \sigma^2 \sim \mathcal{N}\left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1} \right)$$

Variance Update (Both)

$$\sigma^2 | \boldsymbol{\beta} \sim \text{Inv-Gamma}\left(\frac{n}{2}, \frac{1}{2}(y - X\boldsymbol{\beta})^T (y - X\boldsymbol{\beta}) \right)$$

Task 18.1: Implementation

Implement the following in Python:

1. **Run Standard Gibbs:** Update $\beta_0 \to \beta_1 \to \beta_2 \to \sigma^2$ (One by one).
2. **Run Blocked Gibbs:** Update $[\beta_0, \beta_1, \beta_2] \to \sigma^2$ (In groups).

Hyperparameters

$N_{iter} = 10{,}000$, Burn-in $= 1{,}000$, Thinning $= 1$

Task 18.2: Trace Plots and ACF

For both samplers, generate:

1. **Trace Plots**: Plot the value of β_1 against iteration number
2. **Autocorrelation Function (ACF)**: Compute the correlation at various lags and plot them against lag k:

$$\rho(k) = \frac{\mathrm{Cov}(x_t, x_{t+k})}{\mathrm{Var}(x_t)}$$

Observation

Compare the "hairy caterpillar" appearance of the trace plots between the two methods.

Task 18.3: Convergence Diagnostics

1. Run 3 independent chains with different initializations
2. Compute the **Gelman-Rubin statistic ($\hat{R}$)**:

$$\hat{R} = \sqrt{\frac{\widehat{\mathrm{Var}}(\theta|y)}{W}}$$

Requirement

Confirm if $\hat{R} < 1.1$ for all parameters.

The Target: The Rosenbrock (Banana) Density

A "bad" density is one with high curvature or narrow ridges, where standard samplers get "stuck."

The Banana Function

$$f(x_1, x_2) \propto \exp\left(-(1 - x_1)^2 - 100(x_2 - x_1^2)^2\right)$$

- The mode is at $(1, 1)$.
- The density follows a narrow, parabolic ridge.
- **Challenge:** High non-linear correlation. A fixed proposal width in MH will struggle to turn the "corners."

Implementation 1: Metropolis-Hastings (MH)

We use a fixed-width Symmetric Gaussian proposal:
$q(\mathbf{x}^*|\mathbf{x}^{(t)}) = \mathcal{N}(\mathbf{x}^{(t)}, \sigma^2\mathbf{I})$.

The MH Loop Logic

For each iteration t:

1. **Propose**: Generate $\mathbf{x}^* = \mathbf{x}^{(t)} + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I})$.
2. **Acceptance Ratio**: Calculate $r = \frac{f(\mathbf{x}^*)}{f(\mathbf{x}^{(t)})}$.
3. **Decision**:
   - Generate $u \sim \text{Uniform}(0, 1)$.
   - If $u < r$: Set $\mathbf{x}^{(t+1)} = \mathbf{x}^*$.
   - Else: Set $\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)}$ (**Keep the old value**).

Implementation 2: Component-wise Slice Sampling

Slice sampling updates one dimension at a time (x_1, then x_2) by sampling from a horizontal "slice" of the density.

Step-by-Step for each Dimension ($d \in \{1, 2\}$)

1. **Vertical Slice**: Draw $y \sim \text{Uniform}(0, f(\mathbf{x}_{curr}))$.
2. **Stepping Out**: Start with interval $[L, R]$ of width w around x_d.
    - While $f(L, x_{-d}) > y$: $L = L - w$.
    - While $f(R, x_{-d}) > y$: $R = R + w$.
3. **Shrinkage**:
    - Sample $x_{prop} \sim \text{Uniform}(L, R)$.
    - If $f(x_{prop}, x_{-d}) > y$: Accept $x_d^{(t+1)} = x_{prop}$.
    - Else: Shrink the interval (L or R) to x_{prop} and re-sample.

Task 19.1: Simulation and Tuning

1. **Implement MH**: Run the sampler with three different proposal widths:
   - $\sigma^2 = 0.01$ (Tiny steps, high acceptance, slow exploration).
   - $\sigma^2 = 5.0$ (Large steps, low acceptance, "stuck" chain).
   - $\sigma^2 = 0.5$ (Attempt to find the "sweet spot").
2. **Implement Slice**: Run with initial width $w = 1.0$.
3. **Execution**: Run both for $N_{iter} = 10{,}000$.

Observation

Record the **Acceptance Rate** for MH. Note that Slice Sampling technically accepts a new point every iteration!

Task 19.2: Diagnostic Comparison

Compare the "Best" MH run against the Slice Sampler:

1. **Scatter Plots**: Plot the samples over the density contours. Which sampler fills out the "arms" of the banana?
2. **Autocorrelation (ACF)**: Plot the ACF for x_2. Which method shows the correlation dropping to zero faster?
3. **Trace Plots**: Observe the mixing. Does the MH chain stay in one part of the banana for too long?

Analysis Question

Why is the "fixed" proposal width of MH a disadvantage when the density has regions of both high and low curvature?